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Key Points:
- We investigate ecohydrological separation with weekly bulk soil and seasonal xylem isotopic observations using a hydrologic model
- We use StorAge Selection functions to compare the age selection of percolation and evapotranspiration along a hillslope
- Ecohydrological separation may be temporally and spatially heterogeneous

Supporting Information:
- Supporting Information SI

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Seasonal and Topographic Variations in Ecohydrological Separation Within a Small, Temperate, Snow-Influenced Catchment

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Abstract The hypothesis of ecohydrological separation (ES) proposes that the water contained in surface soils is not uniformly extracted by root water uptake nor uniformly displaced by infiltration. Rather vegetation selectively removes water held under tension, and water infiltrating wet soil will bypass much of the water-filled pore space. Methodological differences across previous studies have contributed to disagreement concerning the prevalence of ES. We measured stable isotopes of O and H in precipitation, snowpack, canopy throughfall, and stream water over a period of 18 months in a temperate catchment. At six locations across a wetness gradient, we sampled bulk soil water isotopes weekly and xylem water of Eastern hemlock and American beech stems seasonally. We used these observations in a soil column model including StorAge Selection functions to estimate the isotopic composition and ages of groundwater recharge and ET. Our findings suggest ES may exist with spatial and temporal heterogeneity. Root water uptake ages possibly vary between Eastern hemlock and American beech, suggesting functional strategies for water uptake may control the presence of ES. Newly infiltrated water bypassing the shallow soil was the most likely explanation for bulk soil isotopic measurements made at upslope locations during the winter and summer seasons, whereas rapid displacement of stored soil water by infiltrated waters was the most likely during the spring and fall seasons. Future research incorporating high temporal frequency soil and plant xylem water isotopic measurements applied to StorAge Selection functions may provide a useful framework for understanding rooting zone isotope dynamics.

1. Introduction

Stable water isotope observations within watersheds have led researchers to suggest the presence of two discernable hydrologic pools, termed ecohydrological separation (i.e., two water worlds hypothesis, TWW; Evaristo et al., 2015; McDonnell, 2014; Brooks et al., 2010). Ecohydrological separation proposes that (1) dry season precipitation becomes tightly bound soil water, ultimately leaving the watershed via evaporation and plant transpiration and (2) wet season precipitation (and snowmelt) becomes runoff, mobile soil water, or groundwater recharge, leaving the watershed via streamflow. Ecohydrological separation describes hydraulically matrix bound water as immobile, available only to evaporation or transpiration. This hypothesis is broadly supported by a global evaporation fractionation signal within shallow soil water and plant xylem and a contrasting lack of such signal within groundwater and streamflow (e.g., Evaristo et al., 2015).

Previous efforts to understand ecohydrological separation have centered on (1) separation of soils and plant xylem from that of stream water in the dual water isotope (H and O) space (e.g., Brooks et al., 2010; Evaristo et al., 2015), (2) the degree to which antecedent soil water mixes with newly infiltrated precipitation (e.g., Sprenger et al., 2016; Berry et al., 2017; Knighton et al., 2017), or (3) differences in the ages of rooting zone percolate and water taken up by roots (Evaristo et al., 2019). Methodological differences across studies have possibly contributed to disagreement concerning the interpretation of isotopic evidence and, subsequently, disagreement on the prevalence of ecohydrological separation (Luo et al., 2019).

Support for ecohydrological separation has been observed in tropical, subtropical, and Mediterranean climate regions through separation of streams and tree stored water samples in the dual isotope space (e.g., Brooks et al., 2010; Evaristo et al., 2015). Many watershed-scale experiments in seasonal climates have demonstrated at least some evidence of ecohydrological separation (Brooks et al., 2010; Goldsmith et al., 2012; Gierke et al., 2016; Evaristo et al., 2016; Bowling et al., 2017; Oerter & Bowen, 2017) and so did meta-analysis at the global scale (Evaristo et al., 2015). In contrast, higher latitude catchments with reduced seasonality provided less compelling evidence for ecohydrological separation (e.g., Geris et al., 2015, 2017). McCutcheon et al. (2017) demonstrated evaporation fractionation of soil water and plant xylem in a cool semiarid watershed, though they concluded that the evaporative fractionation signal may not be indicative of ecohydrological separation.

Analyses explicitly considering both the isotopic composition of soil water, δ_{SOILS}, and temporal changes in isotopic composition (e.g., $\frac{d\ \delta_{SOILS}}{dt}$) have frequently yielded conclusions in conflict with ecohydrological separation in that they support vadose zone mixing of antecedent soil water and newly infiltrated water (Sprenger et al., 2016; Sprenger et al., 2017; Knighton, Saia, et al., 2017; Brinkmann et al., 2018) or the more extreme case of piston flow (Tan et al., 2017; Yang & Fu, 2017). Brinkmann et al. (2018), Sprenger et al. (2016), and Mueller et al. (2014) concluded that shallow soil water was continuously mixed with incoming precipitation and transported to deeper soils through analyses of soil water isotopes within mechanistic modeling frameworks. Knighton, Saia, et al. (2017) used observations from a temperate catchment and a mechanistic hydrologic model to estimate the degree of ecohydrological separation in the Northeast United States. Their results suggested that most of the infiltrated water followed TWW behavior potentially via preferential flow in macropores as described by Thomas et al. (2013); however, approximately 30% of groundwater recharge was composed of displaced antecedent soil water.

Recent research has highlighted the possibility that ecohydrological separation varies with catchment wetness. Meta-analysis of the sources of plant root water uptake (RWU) suggests the contribution of groundwater to transpiration is likely a function of soil moisture availability (Barbeta & Peñuelas, 2017; Evaristo & McDonnell, 2017). Vargas et al. (2017) and Stumpp and Maloszewski (2010) performed soil column experiments examining the displacement of "old" isotopically labeled soil water by young infiltrated water under differing soil moisture conditions. Under saturated conditions Vargas et al. (2017) observed significant antecedent soil water displacement, in contrast with the expected TWW behavior; whereas the column observations of Stumpp and Maloszewski (2010) forced with natural meteorological boundary conditions supported TWW. Sprenger, Tetzlaff, Buttle, Laudon, and Soulsby (2018) adequately reproduced bulk and mobile soil water through numerical representation of a water vapor exchange between mobile and immobile soil water, where the degree of mixing is a function on the relative volumes of mobile and immobile soil water. Zhao et al. (2018), Oerter and Bowen (2017), and Herve-Fernandez et al. (2016) similarly demonstrated through detailed temporal monitoring of soils that ecohydrological separation may not hold under wet conditions. Evaristo et al. (2019) evaluated ecohydrological separation within a model tropical ecosystem by comparing the distributions of water ages (i.e., the transit time distribution, TTD) for percolate and transpiration, where differences in the TTDS of these fluxes implied ecohydrological separation in time. They found that ecohydrological separation was evident at a simulated transition from drought to rewetted soil conditions and less evident during a simulated drought. There is evidence that TTDs of stream discharge are a function of wetness at the catchment-scale, termed the "inverse storage effect" (e.g., Harman, 2015; Kim et al., 2016; Pangle et al., 2017). It is possible that water fluxes of percolate and RWU, occurring at smaller spatial scales, exhibit similar wetness-conditioned age selection.

We aim to determine ecohydrological separation exhibits a spatial and temporal heterogeneity within the rooting zone. StorAge Selection (SAS) functions describe the relationship between the ages of water within a control volume (e.g., the rooting zone) and the age of a flux leaving that control volume (e.g., RWU and percolate; Rinaldo et al., 2015). We frame our research questions in the context of parameterized SAS functions which allow for comparison of the ages of percolation and plant RWU. We ask the following questions with this research:

1. Is there evidence of preferential vertical percolation of newly infiltrated water along a hillslope for all seasons?
2. Is there a clear separation of the age selection functions of percolated and evapotranspired water along a hillslope for all seasons?

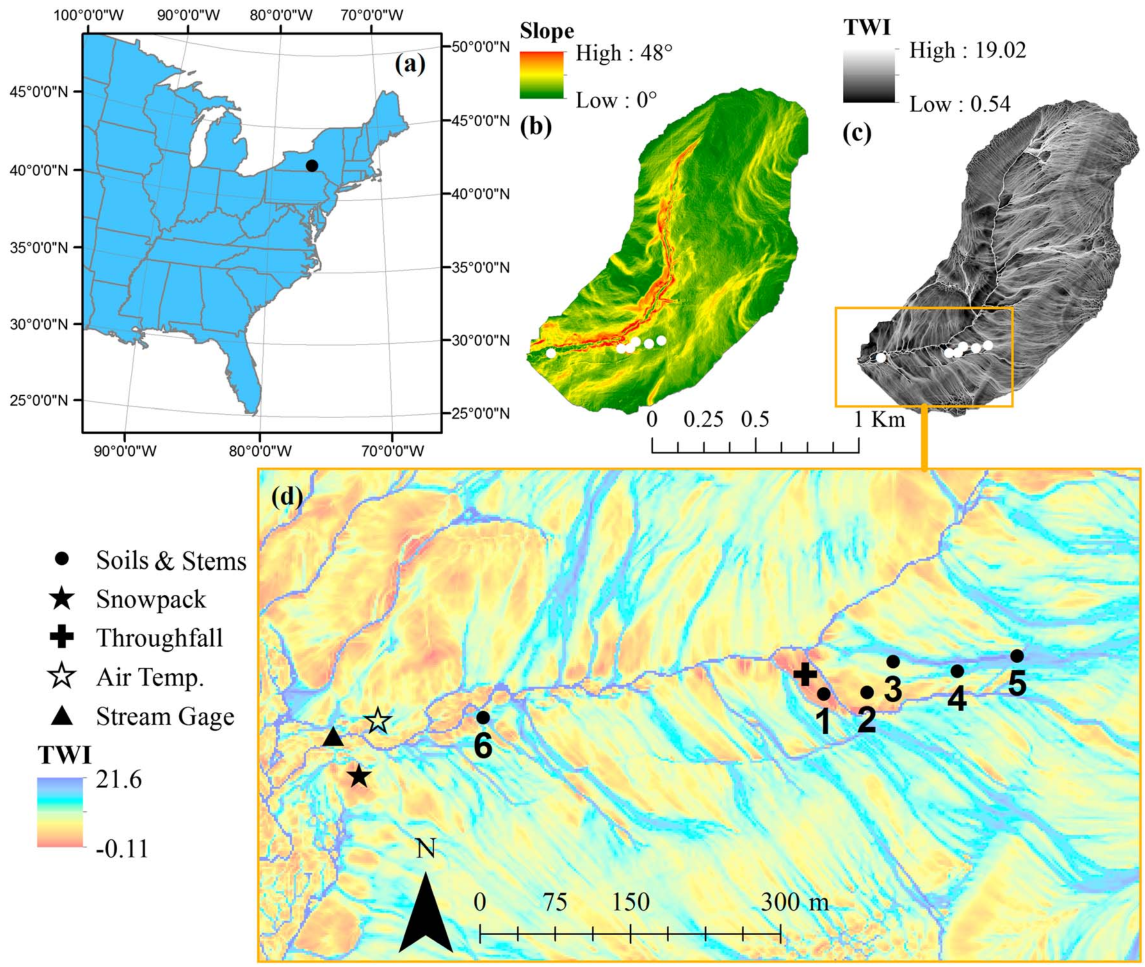

Figure 1. Study catchment (a) overview, (b) land surface slope, (c) TWI, and (d) sampling and instrumentation locations. TWI = topographic wetness index.

2. Methodology

2.1. Study Site Description

The study was conducted in a forested 121-ha watershed within central New York, USA (42.41°, −76.37°) over an 18-month period from 8 January 2017 to 25 June 2018. The watershed contains a second order intermittent stream which is tributary to Six Mile Creek and ultimately Cayuga Lake. Land surface elevation throughout the watershed ranges from 390- to 590-m mean sea level (MSL). Watershed slopes range between 0° and 48° (Figure 1b). The highly dendritic patterns of the surface waters lead to a broad range of topographic wetness index (as calculated in Buchanan et al., 2014) values within the watershed (Figure 1c).

We selected six soil and plant stem sampling sites across a topographic wetness index (TWI) gradient (Figure 1d). Shallow soils of upland sites are primarily Mardin series channery silt loam (saturated hydraulic conductivity ~4 mm/hr) and sandy loam (saturated hydraulic conductivity >10 mm/hr) with a highly organic 5-cm surface layer and increasing gravel content with depth (Natural Resources Conservation Service (NRCS), 2019). Riparian areas are composed the same soil formation, with a low organic content and a consistently higher gravel content (NRCS, 2019; detailed soil properties are presented in the supporting information). Below 50 cm, the subsurface is primarily weathered bedrock. A shallow confining layer of siltstone at an average depth of 1 to 1.5 m (NRCS, 2019) results in frequent saturation-excess overland flow such that the TWI is a strong predictor of shallow soil water content (Buchanan et al., 2014). We estimate soils field capacity, Θ_{FC}, and wilting point, Θ_{WP}, to be approximately 0.27 and 0.13, respectively, with pedotransfer functions using soil textures and organic content of the top 10 cm (Saxton & Rawls, 2006). NRCS (2019) reports regional available water capacity (Θ_{AWC}) as varied between 0.17 and 0.25, providing coarse validation of the pedotransfer derived Θ_{FC} and Θ_{WP}.

The watershed is forested with approximately 65% canopy cover. We conducted a woody plant survey on 9 July 2017 at each sampling site. The catchment canopy was composed of American Beech (*Fagus grandifolia*), Basswood (*Tilia americana*), Eastern hemlock (*Tsuga canadensis*), northern red oak (*Quercus rubra*), red maple (*Acer rubrum*), and sugar maple (*Acer saccharum*). The catchment understory is composed of the woody trees and shrubs American hophornbeam (*Ostrya virginiana*), American beech, American hornbream (*Carpinus caroliniana*), bigtooth aspen (*Populus grandidentata*), black birch (*Betula lenta*), cucumber tree (*Magnolia acuminata*), eastern white pine, (*Pinus strobus*), mountain camellia (*Stewartia ovata*), striped maple (*Acer pensylvanicum*), tulip poplar (*Liriodendron tulipifera*), and white ash (*Fraxinus americana*), as well as various ferns and grasses. American beech (AB) and Eastern hemlock (EH) are common to each soil sampling location and account for 45% and 30% by count of all trees and shrubs surveyed in the watershed respectively. AB was identified in both tree (diameter at breast (DAB) > 10 cm) and shrub form, 70% and 30%, respectively, by count; whereas EH was observed near each sampling location only as a well-established tree. Within this catchment mature (DAB > 10 cm), AB and EH occur with densities of 0.12 and 0.31 m^{-2}, respectively. We note that the six chosen sampling locations have a similar plant species composition despite variations in local soil wetness.

2.2. Hydrologic and Meteorological Data Collection

During rainfall events, hourly gross precipitation (GP) intensity was measured with a 200-mm diameter tipping bucket rain gage (TR-525-S-U, Texas Electronic; precision = 0.2 mm, accuracy = ±3%) installed approximately 0.5-km west of the watershed outlet at an elevation of 436-m MSL, approximately 1-m above ground elevation in an open field. Snow event hourly GP was measured with a weighing gage (Fischer & Porter; precision = 2.5 mm) at the Game Farm Road long-term meteorological station (Natural Resources Conservation Service (NRCS), 2017) approximately 10 km from the watershed outlet at elevation 290-m MSL. Ambient air temperature was measured with a temperature sensor (U20-001-04, HOBO) at an interval of 10 min near the watershed outlet (resolution = 0.1 °C and accuracy = 0.037 °C).

Vertical snowpack cores were collected with a 3-cm diameter core sampler at a weekly interval near the watershed outlet (Figure 1). Snow Water Equivalent (SWE) was calculated by weighing collected snowpack cores. Weekly unsaturated zone volumetric water content (θ, where VWC indicates the water equivalent depth) measurements were taken with a 12-cm time domain reflectivity sampler (CS658, Campbell Scientific; resolution = 0.05% and accuracy = 3%). Each θ observation presented is the arithmetic mean of five measurements taken within a 1 m^2 area as in Buchanan et al. (2014).

2.3. Isotopic Data Collection and Analysis

Weekly samples of accumulated GP ($n = 75$), weekly accumulated throughfall (TF; April to November 2017; $n = 21$), weekly snowpack ($n = 21$), weekly streamflow ($n = 69$), weekly shallow bulk soil water from 5 to 10 cm ($n = 316$), seasonal bulk soil water depth profiles ($n = 122$), and seasonal plant stem samples of AB ($n = 101$) and EH ($n = 107$) were analyzed for δ^2H and $\delta^{18}O$.

Weekly volume-weighted GP isotopic signature, δ_{GP}, was measured using a 15-cm diameter collection jar installed adjacent to the tipping bucket rain gage throughout the experiment duration. As the study site was forested with a dense canopy, it was possible that δ_{GP} would not be representative of TF isotopic composition, δ_{TF}, reaching the forest floor (e.g., Liu et al., 2015; Soulsby et al., 2017). We collected precipitation TF weekly during the growing season (April to November 2017) near soil sampling site 1 beneath a mixed canopy of EH, AB, and northern red oak. Approximately, 50 ml of streamflow and snowpack (when present) were collected weekly at the watershed outlet for isotopic analysis. Snow core samples were representative of the average snowpack water.

Shallow soil samples were collected weekly from 8 January 2017 to 29 December 2017 with a hand pick from a depth of 5 to 10 cm. Approximately, 100-ml samples of soil were collected at each of the six sampling locations. The sampling location was varied at each site within a 1-m^2 area to prevent previous samples from interfering with shallow soil water mixing while drawing from a consistent TWI landscape position.

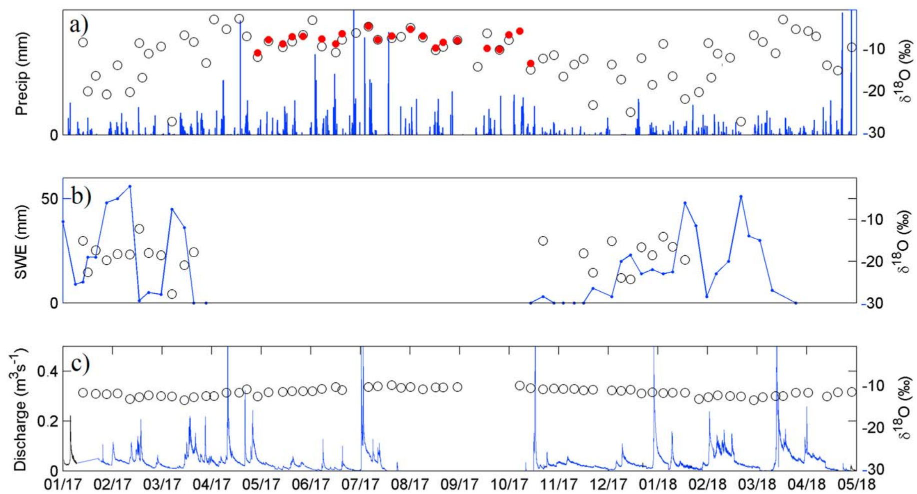

Figure 2. Observed (a) precipitation volume (blue), isotopic composition (open), and canopy throughfall (red), (b) snow water equivalent (blue) and snowpack isotopic composition (open), and (c) streamflow (blue) and stream isotopic composition (open). SWE = Snow Water Equivalent.

Seasonal soil cores and plant stem samples of AB and EH were collected at each site on 7 August 2017, 12 November 2017, and 5 June 2018 from approximately 14:00 to 16:00 on each day. Soil samples were collected at 5-cm increments until 5-cm depth or auger refusal from weathered bedrock was encountered. In two cases, samples from the soil core were not analyzed because of a prohibitively high gravel content (site 6, August, and site 1, June). During each seasonal collection, six 5-cm stem samples of each species were collected at each sampling location. AB was prevalent at site 2 only as a shrub and therefore only three stem samples were collected from this location during each collection to limit damage to the understory. Individual branches near the base were clipped from the tree and cut down to approximately 5-cm lengths. Stems were sampled at a height of 1.5 m from trees ranging 5- to 15-cm DAB. Properties of sampled trees are presented in the supporting information.

Bulk soil and plant stem water was extracted via the cryogenic vacuum extraction methodology (described in Orlowski et al., 2016). The use of bulk soil water is most appropriate given the application of these measurements within a SAS framework that explicitly tracks both mobile and immobile water (described in section 2.5). Soil samples were thawed at 20 °C for 12 hr prior to running soil water extractions. The cryogenic extraction process has been shown to potentially result in an underestimation of soil water enrichment (Orlowski et al., 2016) where the magnitude of the error has been related to several aspects of the methodology: soil texture, extraction time, vacuum pressure, and temperature differential. To minimize errors, extractions were run to completion for a minimum of 180 min. Stem and soils were maintained at a maximum pressure of 10 kPa for the duration of the extraction. Samples were maintained at 100 °C and the water collection vial at −196 °C.

Extracted soil and stem water samples were analyzed using an off-axis integrated cavity output spectrometer (OA-ICOS; IWA-35EP, Los Gatos Research, Mountain View, CA, USA) coupled with an autosampler (LC PAL, CTC Analytics AG, Zwingen, Switzerland) for liquid sample injections. Concurrent measurements of three water standards spanning the range of natural water were used to calibrate the isotopic data to the Vienna Standard Mean Ocean Water reference scale. Measurement precision was 0.5‰ for δ^2H and 0.1‰ for δ^{18}O. All other samples were analyzed on a Thermo Delta V isotope ratio mass spectrometer interfaced to a Gas Bench II with a measurement precision of 0.5‰ for δ^2H and 0.24‰ for δ^{18}O.

2.4. Catchment Hydrometeorology and Hydrology

Over the course of the study period the watershed received 1,990 mm of GP with 277 mm occurring as snowfall. Approximately, 50% of GP exited the watershed as streamflow and 50% via evapotranspiration (ET). GP volume and isotopic composition varied seasonally (Figure 2a). The pattern of δ_{GP} followed the expected pattern of depleted winter and enriched summer GP (Figure 2a). There was slight enrichment of TF relative to GP; $\overline{\delta_{TF}-\delta_{GP}}$ was 0.7‰ and 7.9‰ for δ^{18}O and δ^2H, respectively, during the 2017 growing season.

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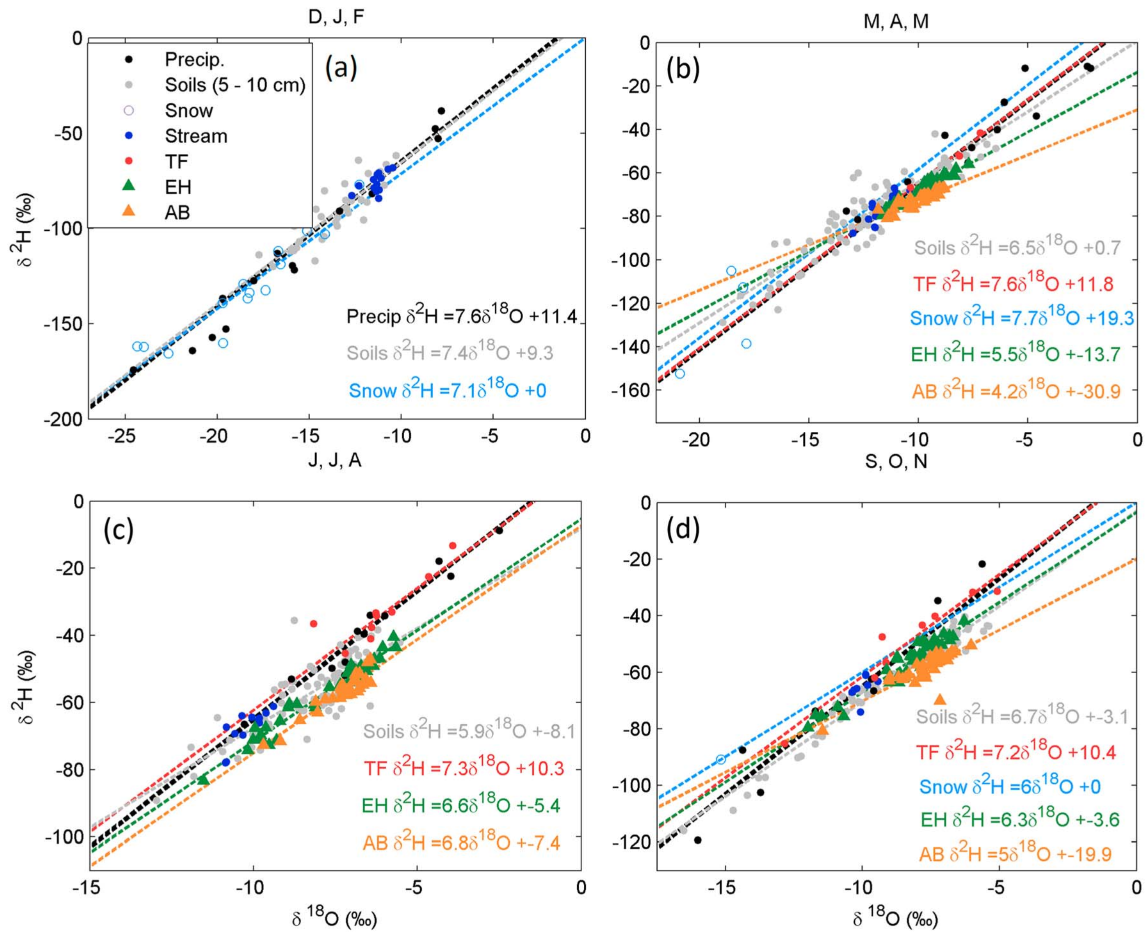

Figure 3. Dual isotope representation of observed precipitation, streamflow, canopy throughfall, soils, EH, and AB for (a) winter (DJF), (b) spring (MAM), (c) summer (JJA), and (d) fall (SON). The precipitation best-fit line is established across all seasons. Tree stem samples collected on 5 June are presented in subfigure b. DJF = December–February; MAM = March–May; JJA = June–August; SON = September–November; TF = throughfall; EH = Eastern hemlock; AB = American beech.

Watershed observations began during the winter season with an established snowpack of approximately 40 mm (Figure 2b). Six distinct periods of snow melt and five periods of snow pack accumulation were observed between January 2017 and June 2018. Newly established snowpack consisted of highly depleted water which enriched throughout periods of melt (Figure 2b). Catchment outflow was seasonal with elevated spring flow following snowmelt (Figure 2c). From August to October 2017, the streambed was dry. During the fall, watershed saturation and streamflow increased following the end of the growing season.

The Local Meteoric Water Line (LMWL) was based on depth-weighted least squares regression of weekly accumulated precipitation δ^2H and δ^{18}O samples. TF, stream water, and snow pack water did not deviate substantially from the LMWL (Figure 3). A strong evaporative fractionation signal for the shallow soils (5–10 cm) was observed only during the JJA and SON seasons (Figures 3c and 3d). Stem samples of EH and AB both show evaporative fractionation, where AB are generally more fractionated than EH across all seasons (Figures 3b–3d).

Shallow soil moisture content (Θ) varied across the six sampling locations (Figure 4; measurements presented only for nonfrozen soils). Above average annual 2017 spring and summer precipitation wetted soils at sites 1, 2, and 3 consistently near Θ_{FC} and sites 4, 5, and 6 near saturation, Θ_n. Site 4 experienced periods of saturation following heavy precipitation and snowmelt but more consistently remained $\Theta_{FC} < \Theta < \Theta_n$. Sites 5 and 6 experienced periods of saturation throughout the fall, winter, and spring. During summer months all sampling locations experienced a drop in Θ, though the effect was most dramatic at the lower TWI sites. Shallow (<15 cm) bulk soil water δ_{SOIL} observations followed a similar pattern to δ_{GP} with depleted winter values and enriched summer values, though the response was varied among sampling

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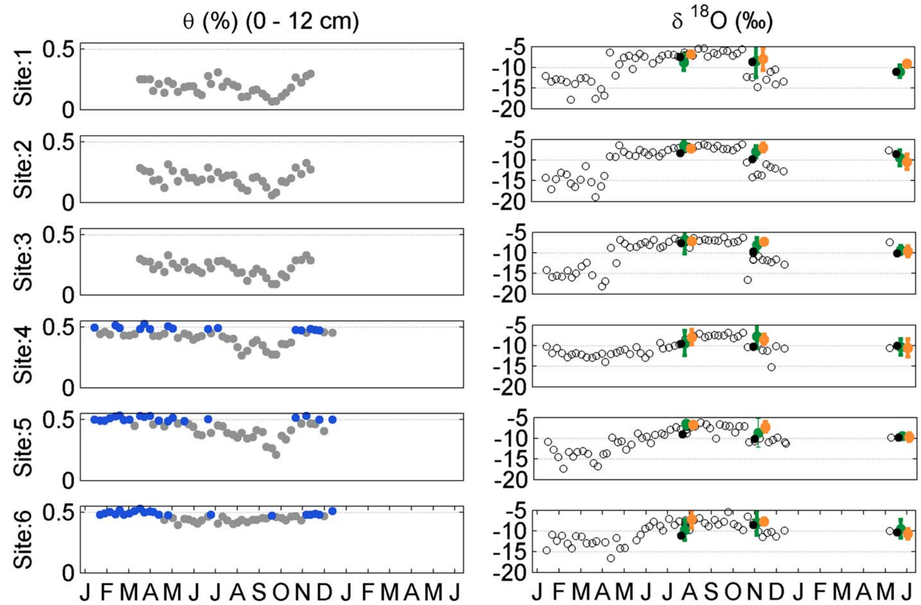

Figure 4. Observed shallow soil volumetric water content (blue dots indicate saturated soils) and bulk soil water 5–10 cm (open dots) the average of soil water isotopic composition for soil depth >15 cm (black dots) and ranges of EH (green) and AB (orange) xylem isotopic composition.

locations (Figure 4). Seasonal samples of EH and AB δ_{XYLEM} showed that stem water tracked closely with shallow δ_{SOIL} during summer and spring, with larger deviations in the fall season possibly suggesting seasonally varied sources of RWU or time lags between the isotopic composition of RWU, δ_{RWU}, and δ_{XYLEM} induced by internal tree water storage.

2.5. Modeling of Shallow Soil Water and Isotopic Transport

We utilized JoFlo, a semi physically based lumped hydrologic model to simulate one-dimensional water mass fluxes within the vadose zone (Figure 5). This model is relatively parsimonious allowing us to limit dimensionality while adequately simulating all relevant ecohydrological processes (Table 2; Knighton, Saia, et al., 2017; Archibald et al., 2014). We computed vadose zone water storage and transport via the Thornthwaite Mather soil water budget (Archibald et al., 2014). Surface runoff was simulated with a modified curve number approach (Archibald et al., 2014). Snowpack accumulation and melt dynamics were simulated with the model of Walter et al. (2005). Snowpack isotopic composition was tracked with the model of Ala-Aho et al. (2017). Daily PET was estimated with the model of Archibald and Walter (2014), which simplifies the land surface energy balance as a function of date of year and daily maximum and minimum land surface air temperatures. Daily PET was then and partitioned between EH and AB based on the relative abundance of each species.

Seasonal isotopic depth profiles of soil water isotopes indicated large variations in δ_{SOILS} of the upper 15 cm, with limited variation below (see supplementary material). We therefore represented the vadose zone as a two-layer model (Figure 5) to allow for both (1) direct comparison to soil isotopic observations and (2) depth variations in RWU. RWU is partitioned between soil layers via the parameter K_{ROOT}, which represents the proportion of PET demand placed on the upper soil layer (Figure 5). The remaining PET demand $(1 - K_{ROOT})$ is apportioned to the lower soil layer. The water storage capacity, V, of each soil layer is defined by the depth of each soil layer, θ_{WP}, and the available water capacity (AWC), a calibrated parameter (Figure 5).

Appropriate soil layer depths were determined from a literature review of AB and EH rooting structures. AB develops roots predominantly within the shallow humic layer of soils (Carpenter, 1974) though may reach a maximum depth of 1.5 m in unrestricted settings (Tubbs & Houston, 1990). Yanai et al. (2008) found that approximately 80% of AB root density existed within the upper 27 cm of soils within a catchment in New Hampshire, USA, suggesting a strong reliance on shallow soil water by AB. The rooting systems of

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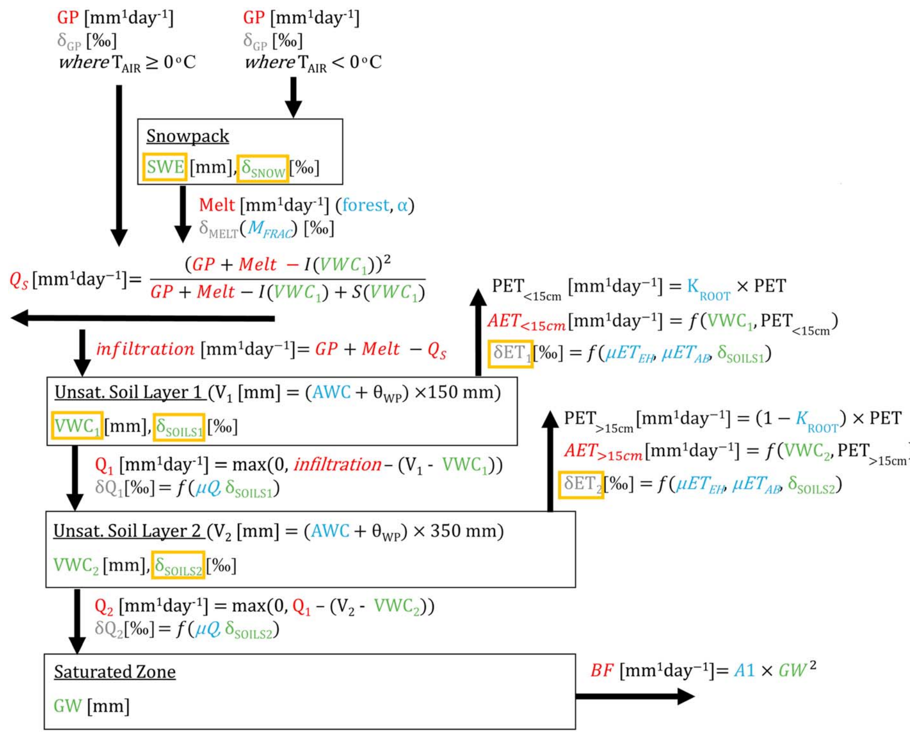

Figure 5. Model schematic showing hydrologic mass fluxes (red), isotopic fluxes (gray), state variables (green), and calibrated hydrologic model parameters (blue). Orange boxes indicate model predictions used in calibration. AET = actual evapotranspiration; AWC = available water capacity; ET = evapotranspiration; GP = gross precipitation; PET = potential evapotranspiration; SWE = Snow Water Equivalent; VWC = volumetric water content.

younger EH trees commonly exist within the upper 20 cm of soils, though at maturity, EH maximum rooting depth is typically restricted by the mean water table elevation (Tubbs & Houston, 1990). Within the nearby (200-km south) Shale Hills CZO with a deep (>2 m) restricting layer, Meinzer et al. (2013) demonstrated that sapflux within EH individuals was strongly correlated to soil water content of the upper 50 cm during the growing season, suggesting a reliance on shallow RWU. Given the shallow confining layer of this study catchment (Natural Resources Conservation Service (NRCS), United States Department of Agriculture, 2019) and evidence of preferential RWU of shallow soil water (<0.5-m depth) by both EH and AB, we adopt the simplifying assumption that EH and AB transpiration draws exclusively from the upper 50 cm within the study catchment. We define the model upper and lower soil layer depths as 150 and 350 mm, respectively (Figure 5).

Isotope transport, storage concentrations, water ages (T) and age-ranked storage (S_T) were simulated with SAS functions (Harman, 2015). The isotope concentration in storage is calculated as $\delta_{soil}(t) = \int_0^\infty \delta_J(\tau) s_T(t{-}\tau, t) d\tau / S(t)$ where $s_T(T,t) = dS_T/dT$, $\delta_J(t)$ is the amount-weighted concentration of precipitation and melt-water inputs, and τ is a time stamp. We defined the age selection of soil water percolation (ωQ), transpiration by EH (ωET_{EH}), and transpiration by AB (ωET_{AB}) with Beta distributions (Figure 5 and Table 1).

The Beta distribution was parameterized in such a way that a single parameter, μ, determines whether a flux is derived from the oldest water in storage ($\mu = 1$), youngest water ($\mu = 0$), or some mixture of all water ages ($0 < \mu < 1$). SAS parameters defined for percolation (μQ, describing percolation from both soil layers), transpiration by EH (μET_{EH}), and transpiration by AB (μET_{AB}) were estimated for each location and season in order to test the research hypothesis that ecohydrological separation varies throughout a catchment and across seasons. We adopted the simplifying assumption that during the winter months (December–February, DJF), $\mu ET_{EH} = \mu ET_{AB} = 0.5$ (i.e., random sampling). For this simplification, we assume that the isotopic composition of ET has relatively little influence on the δ_{SOILS} during the period of plant dormancy and low transpiration rates.

Table 1
Hydrologic Model and SAS Model Description

Flux	Description	References
PET (mm/day)	Daily potential evapotranspiration (PET) is estimated using an energy-based Penman-Monteith model incorporating regional approximations. PET is solved with date of year, latitude, and daily minimum and maximum air temperatures.	Archibald and Walter (2014)
ΔSWE (mm/day)	Snow Water Equivalent (SWE) accumulation and melt (ΔSWE) is solved with the land surface energy and mass balance snow budget estimated from date of year, latitude, and daily minimum and maximum air temperatures.	Walter (2005)
δ_{MELT} (‰)	Snowmelt isotopic composition (δ_{MELT}) was estimated using an empirical approach which simulates time-dependent fractionation of the standing snowpack, where the isotopic composition of melt is determined by an empirical parameter, M_{FRAC}.	Ala-aho et al. (2017)
Q_S and Infiltration (mm/day)	Surface runoff (Q_S) was estimated with the empirical Curve Number reconceptualized to represent saturation excess surface runoff. This model partitions GP and snowmelt into Q_S and infiltration as a function initial abstraction, I, and storage, S, both functions of the upper soil layer water content.	Archibald et al. (2014)
Unsaturated zone water content (mm) and AET (mm/day)	The unsaturated zone water mass balance and daily AET of both unsaturated zone layers is solved with the Thornthwaite-Mather (TM) budget. The TM model is conditioned on the assumption that AET is linearly related to soil water content.	Archibald et al. (2014)
Recharge ($Q_{1,2}$) (mm/day)	Daily recharge from soil layer n, Q_n, is the volume of water inputs to a soil layer in excess of V_n after accounting for AET losses. This simplified model assumes all percolate leaves the soil layer within one time step (1 day).	Archibald et al. (2014), Knighton, Saia, et al. (2017)
BF (mm/day)	Soil column groundwater outflow (BF) is solved as a function of the saturated zone water content (GW) and the recession coefficient A1.	Archibald et al. (2014)
δQ_1, δQ_2 (‰)	Isotopic composition of percolate from each soil layer is solved via the SAS function: $\omega Q(S_T,t) = \frac{(S_T/AW_t)^{\alpha-1}(1-(S_T/AW_t))^{\beta-1}}{\frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}}$ where $\alpha = \sqrt{\frac{\mu Q}{1-\mu Q}}$, $\beta = \sqrt{\frac{1-\mu Q}{\mu Q}}$	Harman (2015)
δET_{EH1}, δET_{AB2} (‰)	Isotopic composition of transpiration fluxes of EH and AB is solved via the SAS function: $\omega ET(S_T,t) = \frac{(S_T/AW_t)^{\alpha-1}(1-(S_T/AW_t))^{\beta-1}}{\frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}}$ where $\alpha = \sqrt{\frac{\mu ET}{1-\mu ET}}$, $\beta = \sqrt{\frac{1-\mu ET}{\mu ET}}$	Harman (2015)

We simulated groundwater (GW) volume and outflow (BF) from each soil column via the recession constant, A1 (Figure 5; Archibald et al., 2014). During periods of rising groundwater, soils at a lower elevation than the groundwater were simulated as wetted to porosity, $\Theta_n = 0.5$. The composite soil water content of each layer, Θ, was then solved as the sum of water contained in the unsaturated (soils above the groundwater elevation) and saturated (soils below the groundwater elevation) portions of the layer to allow for direct calibration to measurements of shallow soil saturation (Figure 4). We assume no isotopic mixing between groundwater and shallow soil water, such that a rising water table does not influence δ_{SOILS}. This simplification is necessary as groundwater isotopic composition is dependent on upstream groundwater recharge and mixing, processes which occur beyond the extent of the model domain. As is demonstrated in the supplemental material (Figures S7–S9), these simplifications do not adversely impact model performance of shallow soil water content or isotopic composition.

The model was run at a daily time step and forced with daily GP, daily δ_{GP} (within each week we use a uniform daily δ_{GP} estimated from the weekly average observed value; δ_{TF} values were used April to November 2017), and daily minimum and maximum air temperatures. The model state variables were initialized by preceding each simulation with a 2-year spin-up utilizing the 2017 forcing data repeated twice. All isotopic pools (SWE and the upper and lower soil layers) were initialized to −15‰ for each simulation, where the spin-up was sufficiently long to prevent influence of initial isotopic conditions of the snowpack and both soil layers.

We used a Bayesian approach (equation (1)) to derive joint posterior pdfs of model parameters, $f_\Phi(\phi \mid y)$, given hydrologic observations, y, prior estimates of parameter distributions, $f_\Phi(\phi)$, and a likelihood function, $\ell(\phi \mid y)$. Hydrologic model parameters, ϕ_{HYDRO}, and residual parameters, ϕ_{GL}, are fit simultaneously as ϕ (equation (2)).

$$f_{\Phi}(\phi \mid y) \propto \ell(y \mid \phi) f_{\Phi}(\phi) \quad (1)$$

$$\phi = [\phi_{HYDRO}, \phi_{GL}] \quad (2)$$

We simultaneously fit our model to each soil and stem sampling location (Figure 1) with seven sets of observations, y: SWE, δ_{SNOW}, upper soil layer VWC, δ_{SOIL1}, δ_{SOIL2}, and δ_{XYLEM} of both EH and AB. Residuals were computed between observed and simulated SWE, δ_{SNOW}, VWC_1, δ_{SOIL1}, δ_{SOIL2}, $\delta_{XYLEM\text{-}AB}$, and $\delta_{XYLEM\text{-}EH}$ (model predictions used in calibration shown in Figure 5). We utilize depth averaged values of observed δ_{SOIL2} of the lower soil layer due to limited isotopic variability with depth (see supporting information). We investigated the sensitivity of model simulation of δ_{SOIL1} to the form of ωET, and demonstrated that reproduction of shallow soil isotopic composition is primarily sensitive to ωQ (see supplementary material). To calibrate ωET, measurements of δ_{XYLEM} were compared to the volume weighted isotopic composition of ET from the upper and lower soil layers for both EH and AB. All replicate δ_{XYLEM} measurements were used, generating 18 residuals (nine for AB site 2) to estimate the ωET and GL function parameters at each site and for each species. We note that δ_{XYLEM} samples were collected seasonally, and therefore may not capture rapid changes in RWU. Calibration to isotopic observations was performed with $\delta^{18}O$. We do not employ dual isotopic calibration (i.e., ^{18}O and ^{2}H) as evaporative fractionation signals in soils have been shown to provide a poor constraint on cool season soil water dynamics (e.g., Knighton, Saia, et al., 2017; McCutcheon et al., 2017).

Feasible value ranges for the five hydrologic parameters (presented in the supporting information) were adopted from Knighton, Saia, et al. (2017) and Archibald et al. (2014). The range for AWC was based on regional variations (NRCS, 2019). Ranges for all eleven SAS parameters were set to the maximum possible ranges (i.e., no a priori parameter constraints). The range for M_{FRAC} was based on Ala-aho et al. (2017). The feasible ranges for five GL parameters were adopted from Schoups and Vrugt (2010), where these parameters are fit for each set of observations, yielding 35 GL function parameters in total for each calibration. We sampled $f_{\Phi}(\phi \mid y)$ through 10,000 model parameter evaluations, discarding the first 1,000 samples as the burn-in period. Posterior samples were selected with a Markov Chain Monte Carlo algorithm via the Metropolis-Hastings sampler supplied with the formal Bayesian generalized likelihood function (GL) of Schoups and Vrugt (2010). Simplistic objective functions such as root-mean-square error (RMSE) assume that model residuals are normally distributed and homoscedastic and are not autocorrelated (Schoups & Vrugt, 2010), though may benefit methodologies through reduced model dimensionality and ease of interpretation. The GL function controls for nonnormal, heteroscedastic, and autocorrelated residuals by simultaneously fitting hydrologic parameters with a five-parameter statistical model. We selected the GL function to control for this possibility and to provide a more reliable appraisal of prediction uncertainty. We present RMSE scores for ease of interpretation, though note that RMSE was not used for model parameter fitting.

Model residuals for each set of observed data were not correlated, and we therefore defined the log-likelihood function as follows:

$$\begin{aligned}\mathscr{L}(\phi \mid y) = {} & \mathscr{L}(\phi \mid \delta_{SNOW}) + \mathscr{L}(\phi \mid SWE) + \mathscr{L}(\phi \mid \delta_{SOIL1}) + \mathscr{L}(\phi \mid \delta_{SOIL2}) + \mathscr{L}(\phi \mid VWC) + \mathscr{L}(\phi \mid \delta_{xylem\_EH}) \\ & + \mathscr{L}(\phi \mid \delta_{xylem\_AB})\end{aligned} \quad (3)$$

Differences in the calibrated values of μQ from μET_{EH} and μET_{AB} indicate that the ages of the water selected by percolate and RWU differ and, therefore, supports ecohydrological separation. Clustering of posterior samples around $\mu Q = 0$ indicates preferential percolate by the youngest soil water. Values $0 < \mu Q < 0$ indicate nonzero contributions to percolate of both newly infiltrated and older antecedent soil water, indicating mixing of younger and older water fluxes at the rooting zone outlet. Clustering around $\mu Q = 1$ indicates piston flow conditions.

3. Results

We sampled the multivariate posterior parameter space, consisting of seventeen hydrologic and SAS function parameters and 35 residual parameters, for each soil and stem sampling location (Figure 1) to better understand spatial and seasonal patterns of water age selection of percolation and ET. Figure 6 presents prediction intervals for simulated time series and RMSE scores of SWE, δ_{SNOW}, Θ, δ_{SOIL1}, δ_{SOIL2} and δ_{XYLEM} for

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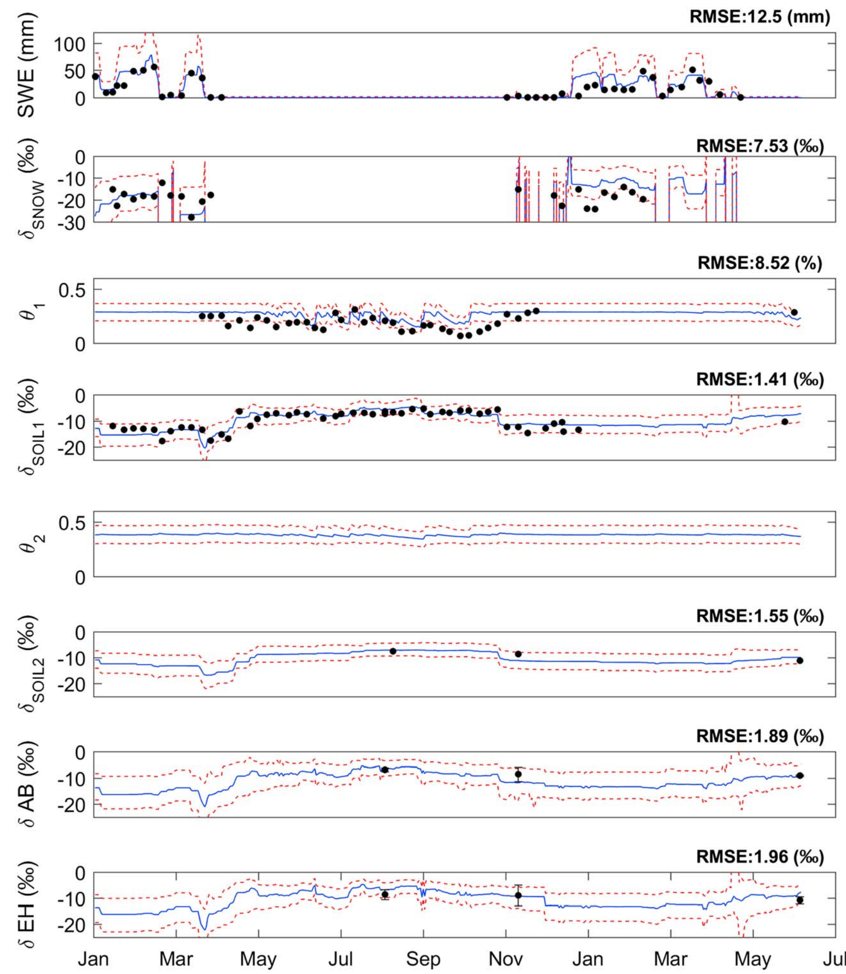

Figure 6. Site 1 (TWI 1.5) snow water equivalent, snowpack isotopic composition, volumetric water content (Θ) of layers 1 and 2, bulk soil isotopic composition for soil layers 1 and 2, and transpired water composition of AB and EH. Red lines indicate 90% confidence intervals of model simulations plus residuals, blue lines indicate median model predictions, and black dots indicate observations. AB = American beech; EH = Eastern hemlock; RMSE = root-mean-square error; SWE = Snow Water Equivalent.

site 1. RMSE scores are calculated from the residuals between observations and the median model prediction. Simulated time series for all other locations are presented in the supporting information along with the marginal posterior histograms of hydrological model parameters and the associated marginal histograms of the GL function parameters for shallow soil isotopic composition and soil water content across all locations.

The GL function parameters for Θ and δ_{SOIL1} (supporting information Figure S16) at all sites demonstrate that these model residuals are nonnormally distributed, heteroscedastic, and in the case of Θ, temporally autocorrelated. These residual properties necessitate the use of the more complex GL function for appropriate model parameter identification. This result highlights the potential for problems stemming from the use of simplistic objective functions based on standard least-squares assumptions (e.g., RMSE, MAE, and NSE) for isotopic model calibration.

The posterior histograms of μQ and μET_{AB} (Figure 7) demonstrate some evidence of ecohydrological separation between percolate and AB RWU during the spring (March–May, MAM) and summer (June–August, JJA) at topographically wetter sites (4, 5, and 6) but less evidence at topographically dry locations (sites 1, 2, and 3). The posteriors of μQ and μET_{EH} provide minimal evidence for ecohydrologic separation between percolate and EH xylem at all locations throughout the spring and summer, with the exception of site 4 during the summer. Separation of μQ from μET_{EH} and μET_{AB} is strongly supported during the fall season where

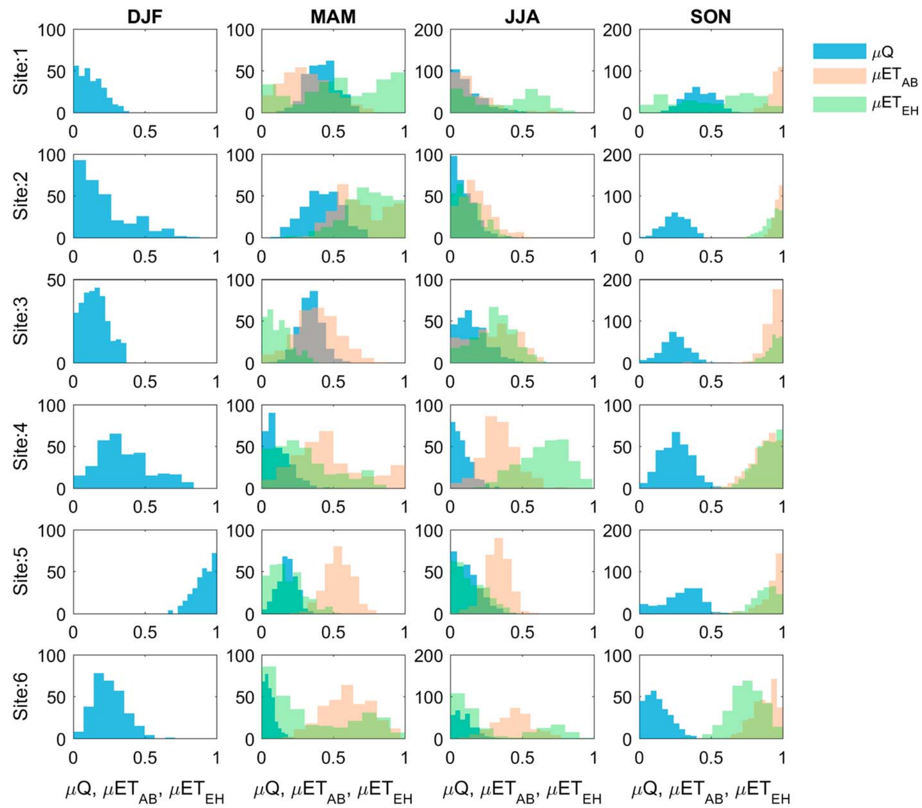

Figure 7. Posterior histograms of seasonal StorAge Selection μ parameters (Table 2) controlling the age ranked selection of antecedent soil water selected during water percolation from the vadose zone. AB = American beech; ET = evapotranspiration; EH = Eastern hemlock; DJF = December–February; MAM = March–May; JJA = June–August; SON = September–November.

percolation draws from younger water than EH and AB (Figure 7), with the exception of EH at site 1. We estimate similarity in the water age selection functions of EH and AB across upland sites 1 and 2 in spring and sites 1, 2, and 3 within the summer and across all sampling locations in the fall. At riparian sampling locations (sites 5 and 6) μET_{AB} and μET_{EH} indicate that AB may access older water than EH through the spring and summer seasons.

The posterior histograms of μQ indicate seasonal and topographic variations in percolate soil water age selection (Figure 7). At upland locations (sites 1,2, and 3), percolate is composed primarily of young water in the winter and summer and older soil water in the spring and fall. For riparian locations with saturated soils, there is limited selection of older soil water during percolation (i.e., μQ ~ 0) through all seasons, possibly due to limited vertical percolation pathways within the saturated extent of the catchment.

4. Discussion

4.1. Separation of Percolate and ET Age Selection Functions

Previous research has defined ecohydrological separation as the distance between plant and soil samples from that of stream water in the dual isotope space (e.g., Brooks et al., 2010). Here we propose an alternate conceptualization of ecohydrological separation where we compare SAS function parameters describing the ages of water selected by percolate and ET across three seasons and six topographic position. The posterior histograms of μQ, μET_{EH} and μET_{AB} across all sites suggest that separation between the ages of percolate and RWU possibly occurs with some seasonality and varies by topographic position (Figure 7).

We observe support for ecohydrological separation in the spring (MAM) and summer seasons (JJA) between percolate and AB at topographically wet locations and strong evidence of separation during the fall season between percolate and both tree species at all locations (except EH at site 1). The finding of strong evidence for ecohydrological separation during the period of soil rewetting is similar to the findings of Evaristo et al. (2019) in a model tropical catchment. This result is in disagreement with several previous studies suggesting ecohydrological separation may not hold during wet seasons (Hervé-Fernández et al., 2016; Oerter & Bowen, 2017; Zhao et al., 2018). Similar to the conclusions of McCutcheon et al. (2017) and Luo et al. (2019), we propose that evidence in the dual isotope space alone does not indicate ecohydrological separation. Through the spring and summer seasons, we observe consistent separation between xylem of both tree species and stream water in the dual isotope space (Figures 3b and 3c) but some similarity in percolate and RWU age selection (Figure 7), particularly for EH.

During the growing season at sampling sites with higher soil water contents (sites 4, 5, and 6 in both spring and summer and site 3 in spring), EH and AB RWU ages possibly differ (Figure 7). This result may be due to differences in the functional strategies of certain trees for accessing stored soil water or possible shallow RWU competition, similar to the conclusions of Knighton et al. (2019) and Brinkmann et al. (2018). Mature individuals of AB have substantially higher specific root lengths than individuals of EH in the top 20 cm of soils (Comas & Eissenstat, 2009), possibly signifying greater dependence on (or a greater competitive advantage for) shallow soil water by AB than EH. Similar to Evaristo et al. (2019) and Brinkmann et al. (2018), our results suggest species-level differences in the ages of RWU, though we also identify a possible interaction between RWU and topographic position.

Berry et al. (2017) discussed the mechanisms of ecohydrological separation within saturated soils. They noted that ecohydrological separation could require RWU of soil water held under high tension despite the availability of mobile water held under lower tension, which would contradict our current understanding of plant physiology. Our results suggest that during the fall season both EH and AB possibly preferentially draw older (and likely less mobile) soil water than the water contributing to percolate, which requires further investigation. Identification of the subsurface sources of RWU has presented some challenges in isotopic studies. Bowling et al. (2017) and Qian et al. (2017) demonstrated uncertainty in the identification of RWU sources in streamside trees with mixing models, though this result is potentially explained by the missing end member of soil water vapor (Oerter et al., 2019). Our field observations do not suggest missing end members for RWU, where the majority of plant stems samples show similar or less evaporative fractionation than bulk soil waters, with the exception of a few individuals of AB in the spring and fall (Figures 3b and 3d). The framework of a numerical soil column model provided additional constraints (e.g., mass balance) beyond the dual isotope approach of Bowling et al. (2017), which potentially assisted in the interpretation of δ_{XYLEM} measurements. Further the use of calibrated SAS functions allows for bulk soil isotopic measurements to be decomposed into mobile and immobile end members by water age, without having measured these end members directly.

Limitations of the experimental design used herein impart some uncertainty on our conclusions which can possibly be remedied in future studies. In particular, the strength of our conclusions pertaining to the water ages of RWU by EH and AB must be tempered as our δ_{XYLEM} sampling was carried out at a low temporal frequency (seasonal), resulting in somewhat poorly constrained posterior histograms for μET_{EH} and μET_{AB} (Figure 7). Complex variations in tree RWU occurring at a higher frequency (as measured by Volkmann et al., 2016 and Evaristo et al., 2019) were not captured by our seasonal sampling design and hydrologic model and, therefore, cannot be adequately described in the posteriors of μET. Further, we included no sampling of δ_{XYLEM} during the winter season as we anticipated low transpiration rates of the coniferous EH and dormancy of the deciduous AB. Instead, we incorporated the open assumption that $\mu ET_{EH} = \mu ET_{AB} = 0.5$ during the winter months, preventing any conclusions concerning age separation of percolate and RWU during the winter.

Field studies employing higher frequency δ_{XYLEM} measurements, water extracted directly from tree roots, and direct measurements of the percolate isotopic composition as in Benettin et al. (2019) may yield alternate conclusions. Additional methodological refinement may be found by coupling SAS functions to hydrologic models capable of more advanced representations of soil water transport (e.g., Sprenger, Tetzlaff, Buttle, Laudon, Leistert, et al., 2018), plant growth dynamics and species-specific functional traits (e.g., Matheny et al., 2017; Kuppel et al., 2018; Knighton et al., 2019), and RWU competition between plant species. Finally, while our findings suggest a temporal and spatial pattern of ecohydryologic separation, we do not attempt to identify the mechanism.

4.2. Seasonal Controls on Unsaturated Zone Percolate

Ecohydrological separation is often described as the existence of hydraulically separate pools of mobile and immobile soil water (e.g., Berry et al., 2017; Knighton, Saia, et al., 2017; Sprenger et al., 2016). Here we review the posterior histograms of μQ across seasons and landscape position.

4.2.1. Seasonal Controls on Percolate at Upslope Locations

During the winter season, δ_{SOILS} at upslope sites 1 to 4 did not show a signal of depleted water in response to infiltration of depleted snowmelt, suggesting preferential bypass of infiltrated water. The marginal posterior histogram of μQ at these sites demonstrates selection of young water for percolate (μQ_{DJF} near 0) was the most likely case during the winter (Figure 7). Watanabe and Kugisaki (2017), Fuss et al. (2016), and Hayashi (2013) described a mechanism where preferential flow paths are developed by freezing of available soil water within smaller pores. Freezing soil water modifies the soil pore size distribution to one dominated by air filled macropores. Broad isotopic observations of groundwater across a similar climate as our study region suggested that groundwater recharge occurs primarily from depleted winter season precipitation, possibly as snowmelt (Jasechko et al., 2017). Our results support a winter recharge hypothesis of snowmelt events occurring over partially frozen soils allowing for preferential groundwater recharge by isotopically depleted snowmelt without displacement of the more enriched frozen soil water.

The March spring snowmelt event resulted in a sustained depletion of δ_{soils} at sites 1 to 3 and a weaker depletion at site 4 with no associated change in Θ. The spring marginal posterior histogram of μQ_{MAM} suggests that snowmelt occurring on thawed soils resulted in substantial flushing of the antecedent soil water (Figure 7). At the watershed scale, Ala-aho et al. (2017) concluded that the spring snowmelt in northern regions is responsible for flushing antecedent soil water from the organic layer into surface waters. One explanation of this observation is the process of soil frost heave, thawing, and settling, structurally alters the soil (Qi et al., 2006), potentially reducing Θ_{FC}. This result could also be explained by the effect of temperature on the soil characteristic curve, where lower temperatures result in reduced soil matric potential (Hayashi, 2013). Alternately, we may consider that the observed snow melt event does displace matrix bound soil water in contrast with previous descriptions of ecohydrological separation (e.g., Berry et al., 2017). The sample mean of the posterior histogram of μQ_{MAM} was inversely proportional to TWI, which may indicate that exchange between mobile and immobile water occurs most readily at upland locations where we anticipate the movement of infiltrated snowmelt to be predominantly vertical.

During the early summer (June and July), soils remained consistently near Θ_{FC} (Figure 4) due to greater than average seasonal precipitation (Figure 2). The marginal posterior of μQ_{JJA} suggests minimal displacement of antecedent soil water during infiltration. In the context of TWW, the observed summer return to preferential selection of newly infiltrated water for percolation could possibly be explained by an evolving structure of watershed soils throughout the study period. Wilson et al. (2017) proposed that macropores tend to form seasonally, with greater preferential flow occurring in drier average locations. Temporary development of macropores during the summer can be facilitated by ecological drivers such as plant roots (Brantley et al., 2017), certain species of cricket (Bailey et al., 2015), and earthworms (Schaik et al., 2014). Weather patterns may similarly explain the difference between spring and summer behavior where higher intensity precipitation events (Figure 2a) can yield a prevalence of preferential flow (Wiekenkamp et al., 2016). Greater magnitude infiltration events similarly reduce vadose zone transit times, which potentially reduces the opportunity for exchange between mobile and immobile water. Such an effect could weaken the water vapor exchange as described by Sprenger et al. (2018).

During the fall season (September, October, and November), bulk soil $\delta^{18}O$ is rapidly depleted and soil water returns to Θ_{fc} across all upland locations with the onset of fall precipitation (Figure 4). The marginal posterior of μQ_{SON} suggests fall precipitation displaces substantial antecedent soil water during the rewetting phase. Though substantial precipitation occurs in excess of late growing season AET, the enriched summer bulk soil water must be displaced to explain the sudden drop in bulk soil isotope values in the fall (Figure 7). The posterior histogram of μQ_{SON} suggests water displacement occurs as partial mixing at upslope locations. The fall increase in soil moisture content possibly alters the hydraulic capacity of the subsurface, closing off preferential flow pathways. Rewetting of dry soils often partially fills and blocks biopores established during the summer season, encouraging matrix flow. Further, rewetting can eliminate macropores formed from shrinkage cracks in clay layers which develop throughout periods of low soil moisture (Nimmo, 2012).

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4.2.2. Seasonal Controls on Percolate at Riparian Locations

The posterior histograms of μQ at sites 5 and 6 are somewhat in contrast with conceptual and numerical descriptions of riparian zones as locations which facilitating subsurface mixing (Birkel et al., 2015; Lessels et al., 2016; Sprenger, Tetzlaff, Buttle, Carey, McNamara, et al., 2018), though we note the posterior distributions show large variance in μQ. An inverse storage effect has been described where streamflow is comprised of a greater proportion of young water during periods of high catchment storage, and older water during low storage (Benettin et al., 2017; Harman, 2015; Kim et al., 2016; Wilusz et al., 2017). Of the two saturated locations, the posterior histograms of μQ within the more consistently saturated location (site 6, $TWI = 12.5$; Figure 7) suggests a greater proportion of young water release than the more moderately saturated location (site 5, $TWI = 10.2$) throughout all seasons. Investigation of time variant TTDs has proposed the inverse storage effect to be caused by rising groundwater, yielding reduced vadose zone storage, truncated flow paths, and increased surface runoff (Kim et al., 2016; Pangle et al., 2017), possibly providing a physical explanation for μQ values near 0.

In a distributed hydrologic modeling framework, Kuppel et al. (2018) simulated mixing of a rising water table and shallow soils such that groundwater fully mixes progressively upward through three vertically stacked soil layers, allowing for some influence of shallow groundwater on the soil isotopic composition. Our model simplification assumed that matrix bound soil water does not mix freely with rising groundwater. This assumption possibly limits our interpretation of μQ in riparian areas, though the reproduction of shallow soil water content and isotopic composition during periods of saturation was reasonable (see supporting information).

4.3. Broader Implications

Within central New York, USA, we anticipate that a future warmer average climate will result in a more ephemeral snowpack, an extended period of spring thaw and reduced spring and summer catchment outflow (Knighton et al., 2017; Knighton et al., 2017). While climate-driven alterations of the hydrologic cycle are often examined within the context of regional shifts in hydrologic extremes (floods and drought), our conclusions contribute to a growing body of research which identifies external modification of TTDs via changing climate forcing (e.g., Wilusz et al., 2017) as an environmental concern. Multiple periods of freeze and thaw of shallow soils possibly have the effect of mobilizing matrix bound soil water and by extension any dissolved constituents. For example, managed agricultural lands could experience spring mobilization of animal fertilizers spread on upland portions of the watershed to limit nutrient mobility (e.g., Knighton et al., 2017). Similar to Ala-Aho et al. (2018), our research highlights a potential relationship between the mobilization of antecedent soil water and the period of soil thawing which may accelerate carbon release and nutrient cycling in temperate and boreal ecosystems. The representation of ecohydrological separation in hydrologic models can drastically alter our understanding of vadose zone water transport and residence times (Cain et al., 2019). Simulation of nutrient and carbon dynamics could therefore potentially be improved with hydrologic models that account for seasonal and spatial patterns in soil water mixing and RWU.

5. Conclusions

We investigated the presence of ecohydrological separation in a small temperate catchment. Previous definitions of ecohydrological separation have largely been conditioned on the separation of plant and soil samples from that of stream water in the dual isotope space. Here we propose an alternate conceptualization of ecohydrological separation framed within the context of a semiphysically based soil column model coupled with SAS functions which describe the ages of percolate and ET by two common tree species, EH and AB. Ecohydrological separation is defined such that calibration to field observations yields differences in the SAS function values for percolate and transpiration.

Over an 18-month period, we made weekly isotopic measurements of precipitation, canopy throughfall (growing season), snow pack (cool season), stream water, bulk soil water, and plant xylem. We collected standard hydrologic measurements of precipitation, snowpack water equivalent, and shallow soil volumetric water content. We applied these observations to a semiphysically based hydrologic model coupled with SAS functions to describe the age ranked selection of these fluxes. A Bayesian Markov Chain Monte Carlo approach was used to derive posterior SAS parameters to estimate the seasonal water age selection patterns.

Though our findings suggest spatial and temporal patterns of ecohydrological separation, uncertainties must be considered when interpreting these results, particularly with respect to the estimation of the age selection of transpiration by EH and AB. Future studies may improve upon this methodology with higher frequency measurements of plant xylem waters and more physically based hydrologic models capable of advanced representations of plant dynamics and soil water transport to better constrain the posteriors of SAS function parameters. We estimated separation between the ages of percolate and AB transpired waters in the spring and summer at wetter topographic positions but limited separation between percolate and EH. Upland locations showed limited evidence for ecohydrological separation during spring and summer. During the fall season, strong separation between the ages of percolate and both tree species occurs, suggesting ecohydrological separation. At upslope locations during the winter (DJF) and summer (JJA) seasons we found evidence for limited displacement of antecedent water following rainfall events. In contrast, during the spring (MAM) and fall (SON) seasons at upslope locations, we estimated that substantial displacement of stored shallow soil water occurs. At riparian locations preferential recharge was the most likely mechanism during the spring, summer, and fall seasons.

We present a possible conceptual model where ecohydrological separation exists ephemerally, through the summer growing season and fall rewetting phase and most predominantly at riparian locations. This research suggests that studies of ecohydrological separation should specifically consider topographic position, season, and tree species as fixed effects.

Annotation List

A1 = baseflow recession coefficient (days^{-1})
AET = actual evapotranspiration (mm/day^{-1})
AWC = available water capacity (mm)
B = backshift operator (dimensionless)
BF = groundwater contribution to baseflow (mm/day^{-1})
forest = proportion of solar radiation intercepted by canopy cover (dimensionless)
GP = gross precipitation (mm/day^{-1})
GW = groundwater storage (mm)
I = initial abstraction (mm)
K_{ROOT} = proportion of root water uptake from upper soil layer (dimensionless)
ℓ = generalized likelihood function of Schoups and Vrugt (2010)
M_{FRAC} = empirical snowpack isotopic enrichment parameter (dimensionless)
Q_S = surface runoff (mm/day^{-1})
S = catchment storage (mm)
SWE = snowpack water equivalent (mm)
V_n = water storage capacity of soil layer n (mm)
VWC = volumetric water content (mm)
y = hydrologic observation vector
α = base land surface albedo (dimensionless)
$\propto_{EC}$ = SAS evapoconcentration coefficient (dimensionless)
β = SEP distribution kurtosis parameter (dimensionless)
δ_{GP} = isotopic composition of gross precipitation (‰)
δ_{SOIL} = bulk soil isotopic composition (‰)
δ_{TF} = isotopic composition of canopy throughfall (‰)
δ_{RWU} = isotopic composition of root water uptake (‰)
δ_{XYLEM} = stem water isotopic composition (‰)
Θ = volumetric water content (m^3/m^3)
Θ_{AWC} = soil available water content (m^3/m^3)
Θ_{FC} = soil field capacity (m^3/m^3)
Θ_n = soil porosity (m^3/m^3)
Θ_{WP} = soil permanent wilting point (m^3/m^3)

$\mu Q =$ SAS parameter defining the release of antecedent water during percolation out of unsaturated zone (dimensionless)
$\mu ET_{EH} =$ SAS parameter defining the age selection of ET by Eastern Hemlock (dimensionless)
$\mu ET_{AB} =$ SAS parameter defining the age selection of ET by American Beech (dimensionless)
$\xi =$ SEP distribution skew parameter (dimensionless)
$\sigma_0 =$ Error standard deviation intercept (dimensionless)
$\sigma_1 =$ Error standard deviation scalar (dimensionless)
$\mathcal{L} =$ log likelihood function
$\phi =$ joint hydrologic model and GL function parameter vector
$\omega Q =$ age selection function of soil water percolate
$\omega ET =$ age selection function of evapotranspiration

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